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RESULTS

Results

01 · CHI-SQUARE INDEPENDENCE

are two categorical variables related?

Table 1. Contingency (observed [expected] (row% / col%) std. residual)

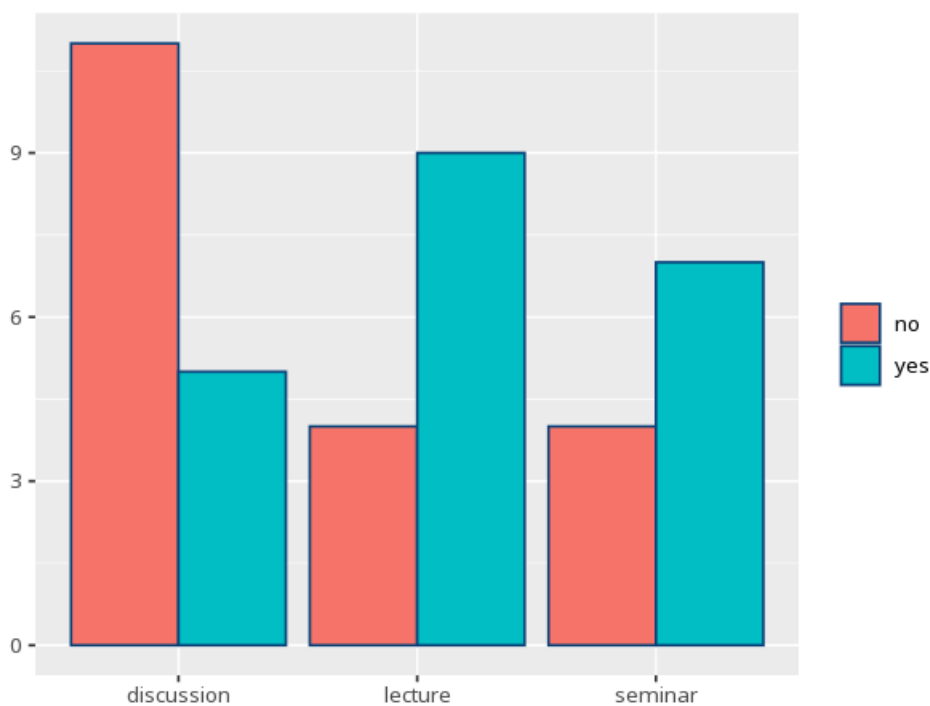
Row \ Column	no		yes	Total
discussion	11 [7.60] (68.8% / 57.9%) $r=2.20$	5 [8.40] (31.3% / 23.8%) $r=-2.20$		16
lecture	4 [6.17] (30.8% / 21.1%) $r=-1.47$	9 [6.83] (69.2% / 42.9%) $r=1.47$		13
seminar	4 [5.22] (36.4% / 21.1%) $r=-0.87$	7 [5.78] (63.6% / 33.3%) $r=0.87$		11
Total	19	21		40

Table 2. Chi-square test

χ^2	df	p	Cramér's V [95% CI]
4.90	2	.086	0.35 [0.00, 1.00]

the contingency table is $r \times c$ — columns expand to the number of categories in the column variable; each cell shows observed [expected] (row% / col%) and its standardized residual r (`chisq.test()`\$stdres), with a warning (suggesting Fisher's exact) if expected counts are too small. A standardized residual beyond about ± 1.96 flags a cell that significantly drives the association. For 2×2 tables `chisq.test()` applies Yates' continuity correction to the χ^2 by default (set `correct=FALSE` for the uncorrected χ^2); the standardized residuals are unaffected by the correction.

Figure. Cross-classification



Understanding the numbers

Row \ Column. The row-variable category heading each row of the contingency table. Here, the table has 3 row categories by 2 column categories.

Cell (observed [expected] / %s / residual). Each cell shows the observed count, the expected count under independence in brackets, its row % and column %, and its standardized residual r . A standardized residual (r) beyond about ± 1.96 flags a cell that significantly drives the association; the smallest expected count here is 5.2.

Cell (observed [expected] / %s / residual). Each cell shows the observed count, the expected count under independence in brackets, its row % and column %, and its standardized residual r . As with the first column, every cell here is read the same way, across 2 column categories.

Additional columns. One more column per remaining category of the column variable, each read the same way. Here, the contingency table expands to 2 column categories in total.

Total. The row or column margin - the sum across that row or column. Here, the table's totals sum to $N = 40$.

χ^2 . The size of the discrepancy between observed and expected counts if the two variables were truly independent. Here, $\chi^2(2, N=40) = 4.90$.

df. Degrees of freedom - $(\text{rows} - 1) \times (\text{columns} - 1)$ for this contingency table. Here, $df = 2$.

p. The probability of an association this strong (or stronger) if the two variables were truly independent. Here, $p = .086$ - at or above your significance threshold, no significant association detected.

Cramér's V. The strength of association, on a 0 (none) to 1 (perfect) scale. Here, $V = .35$ [.00, 1.00].

How to read this test

Tests whether two categorical variables are associated. A p below alpha means they are related; Cramér's V gives the strength of association (0–1); each cell's standardized residual (r) beyond about ± 1.96 flags a category combination that significantly drives the association. Valid only when expected counts are mostly ≥ 5 (the app flags this and suggests Fisher's exact) and each case appears once (raw counts, not percentages). Your significance threshold (α) is 0.05.

APA template: A chi-square test of independence gave $\chi^2(2, N=40)=4.90$, $p = .086$, $V=.35$ [.00, 1.00].

Statistical basis: Chi-square test of independence (Pearson, K. (1900). "On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling." *Philosophical Magazine*, 50(302), 157-175.); Cramér's V effect size (w-family; Cohen, 1988) (Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Routledge.). Full references in the exported CITATIONS.txt.

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